CONCEPTUAL SERIES
Azure Data Factory

Name: ________ Cohort/Batch: ________ Date: ________ Score: ________

AZURE DATA FACTORY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

Q1 What is Azure Data Factory and what problem in Data Engineering does it solve? *Threat Model*

Q2 What are the core components or concepts in Azure Data Factory? *Architecture*

Q3 How does Azure Data Factory compare to its main alternatives? *Configuration*

Q4 How do you get started with Azure Data Factory for a new project? *Hardening*

Q5 What configuration options most affect Azure Data Factory's behavior in production? *SSO / IdP*

Q6 How does Azure Data Factory handle reliability and high availability? *Observability*

Q7 What observability does Azure Data Factory provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in Azure Data Factory? *Compliance*

Q9 What security best practices apply when running Azure Data Factory? *Testing*

Q10 What mistakes do teams commonly make when adopting Azure Data Factory? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Azure Data Factory is a tool

Mitigation

Azure Data Factory is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

A2 Azure Data Factory has key building

Primitives

Azure Data Factory has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Azure Data Factory trades off simplicity

Hardening

Azure Data Factory trades off simplicity against feature richness differently than its alternatives. Choose Azure Data Factory when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Azure Data Factory via its

Gotchas

Install Azure Data Factory via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Azure Data Factory provides HA through

Observability

Azure Data Factory provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Azure Data Factory exposes metrics

Automation

Azure Data Factory exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Azure Data Factory with the

Assessment

Run Azure Data Factory with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.